

2.1) najdu n.s. vektor, a $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \perp \begin{pmatrix} 3 \\ 7 \end{pmatrix}$ a $|\begin{pmatrix} 1 \\ 2 \end{pmatrix}| = |\begin{pmatrix} 3 \\ 7 \end{pmatrix}| = 1 \dots$ ON báse

$$\vec{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \vec{b}_2 = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

musi plati: $\vec{b}_1^T A \vec{b}_2 = 0 \dots$ kolonost

$$\left. \begin{aligned} \vec{b}_1^T A \vec{b}_1 &= 1 \\ \vec{b}_2^T A \vec{b}_2 &= 1 \end{aligned} \right\} \text{jednotková "délka"}$$

existuje matice $A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}$

$\rightarrow$ když najdu matice najdu i přísluš

$$(1 \ 2) \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\vec{b}_1^T A \vec{b}_1 = (1 \ 2) \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (a_1 + 2a_2 + 2a_3 + 4a_4)$$

$$\dots a_1 + 2a_2 + 2a_3 + 4a_4 = 1$$

$$\vec{b}_2^T A \vec{b}_2 = (3 \ 7) \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = (3a_1 + 7a_2 + 3a_3 + 7a_4) = (9a_1 + 21a_2 + 21a_3 + 49a_4)$$

$$9a_1 + 21a_2 + 21a_3 + 49a_4 = 1$$

$$\vec{b}_1^T A \vec{b}_2 = (1 \ 2) \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = (a_1 + 2a_2 + 3a_3 + 7a_4) = (3a_1 + 6a_2 + 7a_3 + 14a_4)$$

$$3a_1 + 7a_2 + 6a_3 + 14a_4$$

SCR: $a_1 + 2a_2 + 2a_3 + 4a_4 = 1$
 $9a_1 + 21a_2 + 21a_3 + 49a_4 = 1$
 $3a_1 + 7a_2 + 6a_3 + 14a_4 = 0$

$$\left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 9 & 21 & 21 & 49 & 1 \\ 3 & 7 & 6 & 14 & 0 \end{array} \right) \xrightarrow{(-3)} \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 3 & 3 & 13 & -8 \\ 0 & 1 & 0 & 2 & -3 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 1 & 0 & 2 & -3 \\ 0 & 3 & 3 & 13 & -8 \end{array} \right) \xrightarrow{(-3)} \left(\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 1 \\ 0 & 1 & 0 & 2 & -3 \\ 0 & 0 & 3 & 7 & 1 \end{array} \right)$$

$$\left\{ \begin{pmatrix} 19/3 \\ -3 \\ 4/3 \\ 0 \end{pmatrix} + t_4 \begin{pmatrix} 14/3 \\ -2 \\ -7/3 \\ 1 \end{pmatrix} ; t_4 \in \mathbb{R} \right\}$$

$$\begin{aligned} 3x_3 + 7t_4 &= 1 & x_2 + 2t_4 &= -3 \\ x_3 &= \frac{1-7t_4}{3} & x_2 &= -3-2t_4 \\ x_1 + 2x_2 + 2x_3 + 4t_4 &= 1 \\ x_1 &= \frac{19+14t_4}{3} \end{aligned}$$

symetrická matice: $\begin{pmatrix} 19/3 + t_4 \cdot 14/3 & -3 + t_4(-2) \\ 1/3 + t_4(-7/3) & t_4 \end{pmatrix} ; t_4 \in \mathbb{R}$

~~matice~~

! ALE matice musí být symetrická, což máš tam pro $t_4 = 10$ a pak $A = \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix}$

pro $(x_1 \ x_2) \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \dots \dots \langle \vec{x}, \vec{y} \rangle = \underline{53x_1y_1 - 23x_2y_1 - 23x_1y_2 + 10x_2y_2}$